

Version	v1.0 — MVP
Platform	Android + iOS
Stack	React Native + FastAPI + MongoDB
AI Engine	OpenAI GPT-4o-mini
Market	India (Hindi + English)

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1. Executive Summary

Vocash is a voice-powered, AI-driven expense tracking mobile application built specifically for the Indian market. Users simply speak their expenses in Hindi or English — the app automatically extracts the amount, category, and details using AI, and saves it instantly. No typing, no manual entry, no friction.

Core Problem: Existing expense trackers require manual typing. Indians are highly conversational and voice-first — yet no app supports Hindi voice input for financial tracking.

Core Solution: Speak naturally → AI understands → Auto-categorized expense saved. Powered by OpenAI GPT-4o-mini with a smart daily limit system to control costs.

1.1 Unique Value Proposition

Competitor	Hindi Voice	AI Extraction	India-First	Price
Vocash	Yes	Yes	Yes	Free / Rs. 99/month
Walnut	No	No	Partial	Free
Money Manager	No	No	No	Rs. 300+
Monefy	No	No	No	Free / Paid

2. Product Overview

2.1 App Screens (MVP)

The MVP will consist of five core screens designed for maximum simplicity and usability:

Screen	Purpose	Key Elements
Login / Signup	User authentication	Email, Password, Google Sign-in
Dashboard (Home)	Financial overview	Monthly total, Pie chart, Last 5 transactions
Voice Input	Core feature — add expense	Mic button, Live transcript, Confirm/Edit
Transactions List	View all expenses	Filter by date, category, amount, Search
Profile / Settings	Account & plan management	Plan status, Upgrade button, Daily limit counter

2.2 Categories (Default)

The following 10 categories will be available in MVP. Users can assign custom categories in the Pro version.

Category	Icon	Examples
Food & Dining	Food	Restaurant, Zomato, Swiggy, Chai, Groceries
Transport	Car	Petrol, Uber, Auto, Bus ticket, Metro
Shopping	Bag	Clothes, Amazon, Flipkart, Electronics
Bills & Utilities	Bill	Electricity, Water, Internet, Recharge
Health	Health	Medicine, Doctor, Hospital, Pharmacy
Entertainment	Movie	Netflix, Movies, Games, Concerts
Education	Book	Tuition, Books, Courses, Stationery
Savings & Investment	Piggy bank	SIP, FD, RD, Savings transfer
Family & Personal	Family	Gifts, Salon, Personal care
Other	Other	Miscellaneous expenses

2.3 Free vs Pro Features

Feature	Free	Pro (Rs. 99/month)
Voice expense entry	3 per day	Unlimited
Manual expense entry	Unlimited	Unlimited
Transaction history	30 days	Unlimited
Default categories	10 categories	10 + Custom
Dashboard & charts	Basic	Advanced analytics
In-app advertisements	Yes	No ads
Cloud backup	No	Yes
CSV / PDF export	No	Yes
Budget alerts	No	Yes (push + email)
Family accounts	No	Up to 3 members
Priority support	No	Yes

3. Complete Tech Stack

3.1 Frontend — React Native

Library / Tool	Purpose	Why Chosen
React Native 0.73+	Cross-platform mobile framework	Single codebase for Android + iOS
React Navigation v6	Screen navigation & routing	Industry standard, well maintained
Zustand	Global state management	Simpler than Redux, lightweight
Axios	HTTP client for API calls	Best Promise-based HTTP library
@react-native-voice/voice	Speech-to-text recording	Native mic access, supports Hindi
React Native Chart Kit	Dashboard pie/bar charts	Easy to use, good visuals
React Native SVG	Chart rendering dependency	Required for Chart Kit
React Native Paper	Material Design UI components	Pre-built polished components

Library / Tool	Purpose	Why Chosen
AsyncStorage	Local device storage / cache	Persist tokens and user data locally
React Native Razorpay	Payment gateway integration	India's best payment SDK
React Native Vector Icons	Icon library	1000+ icons, lightweight

3.2 Backend — Python FastAPI

Library / Tool	Purpose	Why Chosen
FastAPI	Web framework	Fastest Python framework, async support, auto docs
Uvicorn	ASGI server	Production-grade server for FastAPI
Motor	Async MongoDB driver	Non-blocking DB operations for performance
PyMongo	MongoDB utility	Sync operations where needed
python-jose	JWT token creation & validation	Secure auth token management
Passlib + Bcrypt	Password hashing	Industry standard security
OpenAI Python SDK	GPT-4o-mini API calls	Official SDK, easy integration
python-dotenv	Environment variable management	Keep API keys secure
Pydantic v2	Data validation & schemas	Built into FastAPI, type-safe
python-multipart	File upload handling	For future receipt photo upload

3.3 Database — MongoDB Atlas

MongoDB Atlas is used as the primary database. Its flexible document model is ideal for expense data that may vary in structure across users and categories.

Collection	Purpose	Key Fields
users	User accounts	id, email, password_hash, is_pro, created_at
expenses	All transactions	id, user_id, amount, category, item, date, source
usage	Daily API limit tracking	user_id, date, extraction_count
budgets	Monthly category budgets (Pro)	user_id, category, limit_amount, month

3.4 AI Integration — OpenAI GPT-4o-mini

Aspect	Detail
Model	GPT-4o-mini (cheapest, fastest, accurate for extraction)
Cost per extraction	~\$0.00006 (approx Rs. 0.005 per request)
Free user daily cost	3 calls x Rs. 0.005 = Rs. 0.015 per user per day
1,000 free users/month	Approx Rs. 450/month total AI cost
Prompt language	Hindi + English mixed input supported
Response format	Strict JSON: amount, category, item, date, notes
Fallback	Manual entry if extraction fails or limit reached

4. System Architecture

4.1 High-Level Architecture

Architecture Type: 3-Tier — Mobile Client → REST API Backend → Cloud Database + External AI

Layer	Technology	Responsibility
Mobile App (Tier 1)	React Native	UI, voice recording, speech-to-text, local cache
API Server (Tier 2)	Python FastAPI	Business logic, auth, AI routing, limit checking
Data Layer (Tier 3)	MongoDB Atlas	Persistent storage of users, expenses, usage data
AI Service (External)	OpenAI GPT-4o-mini	Natural language expense extraction
Payment (External)	Razorpay	Subscription billing, UPI, card payments

4.2 Voice Expense Flow — Step by Step

User presses mic button in the React Native app

Device microphone records audio using `@react-native-voice/voice`

On-device Speech-to-Text converts audio to raw text (e.g., 'Aaj 500 ka petrol bhara')

App sends raw text to FastAPI backend via `POST /expense/extract`

Backend checks user's daily extraction count in MongoDB usage collection

If free user and count ≥ 3 , return 429 error with upgrade prompt

If allowed, call OpenAI GPT-4o-mini with structured extraction prompt

OpenAI returns JSON: `{ amount: 500, category: 'Transport', item: 'Petrol', date: 'today' }`

Backend increments usage count and returns extracted data to app

User sees pre-filled confirmation screen — can edit if needed

On confirm, app sends final data to `POST /expense/save`

Expense saved in MongoDB — dashboard updates instantly

4.3 API Endpoints

Method	Endpoint	Purpose	Auth Required
POST	<code>/auth/register</code>	New user signup	No
POST	<code>/auth/login</code>	Login, get JWT token	No
GET	<code>/auth/me</code>	Get current user info	Yes
POST	<code>/expense/extract</code>	Voice text → AI extraction	Yes
POST	<code>/expense/save</code>	Save extracted/manual expense	Yes
GET	<code>/expense/list</code>	Get all user expenses (paginated)	Yes
PUT	<code>/expense/{id}</code>	Edit an expense	Yes
DELETE	<code>/expense/{id}</code>	Delete an expense	Yes
GET	<code>/dashboard/summary</code>	Monthly totals + category breakdown	Yes

Method	Endpoint	Purpose	Auth Required
GET	/usage/today	Today's extraction count + limit status	Yes
POST	/payment/verify	Razorpay payment verification	Yes
POST	/budget/set	Set monthly budget per category (Pro)	Yes
GET	/budget/status	Budget vs actual spending (Pro)	Yes

5. Development Phases

Phase 1 — Backend Foundation (Week 1-2)

Goal: Working FastAPI server with auth, AI extraction, and usage limits

Task	Details	Priority
Project setup	FastAPI + virtual env + folder structure	P0
MongoDB connection	Motor async driver + Atlas free cluster setup	P0
User auth	JWT register/login with bcrypt password hashing	P0
MongoDB schemas	Users, Expenses, Usage, Budgets collections	P0
OpenAI integration	GPT-4o-mini extraction with Hindi+English prompt	P0
Daily limit logic	3/day free check + increment in usage collection	P0
Expense CRUD APIs	Create, Read, Update, Delete endpoints	P0
Dashboard API	Monthly summary + category-wise breakdown	P1
Swagger docs	Auto-generated — test all APIs at /docs	P1
Environment config	.env setup for all secret keys	P0

Phase 2 — React Native App (Week 3-4)

Goal: Working mobile app with voice input, login, and transaction list on both Android and iOS

Task	Details	Priority
Project init	npx react-native init Vocash + all package installs	P0
Navigation setup	Stack + Bottom Tab navigator with 5 screens	P0
Auth screens	Login + Signup with JWT token storage	P0

Task	Details	Priority
Voice screen	Mic button, real-time transcript, confirm/edit UI	P0
API service layer	Axios instance with auth header interceptor	P0
Transaction list	FlatList with category filter and date sort	P0
Manual entry form	Fallback when voice is not available	P1
Usage counter UI	Show X/3 free uses today in header	P0
Error handling	Network errors, extraction failures, limit hit	P1
Local token storage	AsyncStorage for JWT persistence	P0

Phase 3 — Dashboard + Analytics (Week 5)

Goal: Rich visual dashboard and analytics for users to understand their spending

Task	Details	Priority
Home dashboard	Monthly total card + category pie chart	P0
Recent transactions	Last 5 expenses on home screen	P0
Category breakdown	Spend per category with percentage	P0
Monthly bar chart	Week-wise spending trend this month	P1
Category filter	Tap category to see only those transactions	P1
Date range filter	This week, this month, last month, custom	P1
Budget progress (Pro)	Category budget vs actual with alert	P2
Search transactions	Full-text search by item name or amount	P2

Phase 4 — Monetization + Polish (Week 6)

Goal: Payment integration, ads, upgrade flow, and Play Store submission

Task	Details	Priority
Razorpay integration	Monthly subscription Rs. 99 via UPI/card	P0
Pro upgrade screen	Benefits showcase + payment button	P0
Limit hit UI	Beautiful upgrade prompt at 3rd extraction	P0
Google AdMob	Banner ads on free dashboard	P1
Rewarded ads	Watch ad for 1 extra extraction today	P1
CSV export (Pro)	Download transactions as spreadsheet	P2
Onboarding flow	3-screen intro for new users	P1
Push notifications	Budget alerts + monthly summary	P2

Task	Details	Priority
App icon + splash	Brand identity implementation	P0
Play Store submission	Screenshots, description, APK upload	P0
App Store submission	TestFlight beta first, then production	P1

6. Deployment Architecture

6.1 Local Development Setup

Service	Local Address	Tool
FastAPI Backend	http://localhost:5000	uvicorn main:app --reload
API Documentation	http://localhost:5000/docs	Swagger UI (auto-generated)
MongoDB	Cloud (Atlas free tier)	MongoDB Atlas M0 cluster
React Native Android	Android Emulator	npx react-native run-android
React Native iOS	iOS Simulator	npx react-native run-ios
Android API URL	http://10.0.2.2:5000	Emulator maps localhost here
iOS API URL	http://localhost:5000	Simulator uses real localhost

6.2 Production Deployment

Service	Platform	Plan	Monthly Cost
FastAPI Backend	Render.com	Free (MVP) → Starter \$7/mo	Rs. 0 → Rs. 585
MongoDB Database	MongoDB Atlas	M0 Free → M2 \$9/mo	Rs. 0 → Rs. 750
Domain Name	GoDaddy / Namecheap	.in domain	Rs. 67/mo (Rs. 800/yr)
Android App	Google Play Store	One-time	Rs. 2,100 one-time
iOS App	Apple App Store	Annual	Rs. 8,200/yr (~Rs. 683/mo)
OpenAI API	OpenAI Platform	Pay-per-use	Rs. 300-500/mo (1K users)
Total MVP Cost			Rs. 500-600/month

6.3 Deployment Flow

Backend Deploy Process: Push code to GitHub → Render auto-detects → Builds + deploys automatically → Live URL generated

Create GitHub repository and push FastAPI project

Connect Render.com to GitHub repository

Set environment variables on Render dashboard (OPENAI_KEY, MONGODB_URL, JWT_SECRET)

Render builds and deploys automatically on every git push

Update API_BASE_URL in React Native app to Render production URL

Build React Native APK for Android: npx react-native build-android

Submit APK to Google Play Console with store listing

Build iOS IPA via Xcode Archive for App Store Connect submission

6.4 Environment Variables Required

Variable	Value Description	Where Used
OPENAI_API_KEY	OpenAI platform API key	Backend (FastAPI)
MONGODB_URL	Atlas connection string with credentials	Backend (FastAPI)
JWT_SECRET	Random 32+ char secret for token signing	Backend (FastAPI)
RAZORPAY_KEY_ID	Razorpay dashboard public key	Backend + Frontend
RAZORPAY_KEY_SECRET	Razorpay dashboard secret key	Backend only
API_BASE_URL	Render.com production URL	React Native app
ADMOB_APP_ID	Google AdMob app identifier	React Native app

7. Monetization Strategy

7.1 Revenue Streams

Stream	Model	When to Start	Realistic Monthly (10K users)
Pro Subscription	Rs. 99-149/month (5% convert)	Phase 4	Rs. 50,000 - 75,000
Google AdMob	Banner + rewarded ads on free users	Phase 4	Rs. 15,000 - 25,000
Affiliate — Groww/Zerodha	Rs. 300-500 per investment signup	Month 4-6	Rs. 20,000 - 40,000
Affiliate — Insurance	Rs. 500-2000 per lead (PolicyBazaar)	Month 6+	Rs. 15,000 - 50,000
B2B Business Plan	Rs. 999/month per business account	Month 8+	Rs. 30,000 - 1,00,000

7.2 Revenue Projections

Phase	Total Users	Paid Users (5%)	Monthly Revenue	Monthly Cost	Net Profit
Month 3	500	0 (free)	Rs. 0	Rs. 500	Rs. -500
Month 6	5,000	250	Rs. 25,000	Rs. 3,000	Rs. 22,000
Month 9	20,000	1,000	Rs. 1,00,000	Rs. 10,000	Rs. 90,000
Month 12	50,000	2,500	Rs. 2,75,000	Rs. 25,000	Rs. 2,50,000

8. Go-To-Market Plan

8.1 Phase-wise Growth Strategy

Pre-Launch (Month 1-2)

Post on LinkedIn: 'Building a Hindi voice expense tracker'

Collect 50 beta testers from friends, college groups, WhatsApp

Create Instagram page for Vocash — start posting content

Register domain and set up a simple coming soon landing page

Launch (Month 3)

Product Hunt launch — target 'Product of the Day'

Post on Reddit: r/IndiaInvestments, r/personalfinance, r/india

YouTube Shorts: 30-second demo of voice input working

Quora answers on 'best expense tracker India' — mention Vocash naturally

Growth (Month 4-8)

College community outreach — NIT, IIT, DU Facebook groups

Free 6-month Pro for 20 micro-influencers (10K-100K followers)

App Store Optimization — Hindi keywords, India-specific screenshots

YouTube educational content: 'Budget kaise manage karein' series

Scale (Month 9-12)

Launch B2B business version at Rs. 999/month

Add affiliate banners for Groww, Zerodha, PolicyBazaar

Regional language expansion: Tamil, Telugu, Marathi

Press outreach: YourStory, Inc42, Economic Times Startups

8.2 Free Promotion Channels (Zero Budget)

Channel	Strategy	Expected Reach
LinkedIn	Build in public — weekly progress posts	500-2,000 followers
YouTube Shorts	30-60 sec voice demo videos in Hindi	5,000-50,000 views/video
Reddit India	Helpful posts in finance subreddits	500-5,000 per post
Quora	Answer expense tracking questions with app link	Organic search traffic
WhatsApp Groups	Personal finance + college groups	100-500 per group
Product Hunt	Global launch day — one-time event	500-3,000 installs
Instagram Reels	Before/after content, finance tips	2,000-20,000 per reel
College Communities	Facebook groups + WhatsApp	Direct targeted users

9. Cost Summary — All Phases

9.1 One-Time Costs

Item	Cost	Notes
Google Play Developer Account	Rs. 2,100	One-time lifetime fee
Apple Developer Program	Rs. 8,200/year	Required for App Store
Domain Name (.in)	Rs. 800/year	vocash.in or similar
Total One-Time (Year 1)	Rs. 11,100	Including Apple annual fee

9.2 Monthly Running Costs by Scale

Service	0-1K Users	1K-10K Users	10K-50K Users
Render.com (Backend)	Free	Rs. 585/mo (\$7)	Rs. 2,100/mo (\$25)
MongoDB Atlas	Free	Rs. 750/mo (\$9)	Rs. 4,750/mo (\$57)
OpenAI API	Rs. 225/mo	Rs. 2,250/mo	Rs. 15,000/mo
Domain	Rs. 67/mo	Rs. 67/mo	Rs. 67/mo

Service	0-1K Users	1K-10K Users	10K-50K Users
Miscellaneous	Rs. 0	Rs. 500/mo	Rs. 2,000/mo
Total Monthly Cost	Rs. 300-500	Rs. 4,500-5,000	Rs. 24,000-25,000
Expected Revenue	Rs. 0	Rs. 50,000+	Rs. 2,50,000+
Net Profit/Loss	Rs. -500	Rs. 45,000+	Rs. 2,25,000+

10. Risks & Mitigation

Risk	Likelihood	Impact	Mitigation
Speech-to-text accuracy for Hindi is low	Medium	High	Use Google Cloud STT — best Hindi support. Offer manual edit after extraction.
OpenAI API cost spikes with growth	Medium	Medium	3/day limit already controls this. Upgrade to Pro removes constraint profitably.
App Store rejection (Apple)	Low	High	Follow Apple HIG guidelines strictly. Test with TestFlight before submission.
Low conversion from Free to Pro	Medium	High	Strong upgrade prompts at limit hit. Rewarded ads as middle tier option.
Competitor launches similar Hindi app	Low	Medium	Move fast. First mover advantage. Build community loyalty early.
User data privacy concerns	Low	High	Clear privacy policy. No selling data. Transparent AI usage disclosure.
MongoDB Atlas data loss	Very Low	High	Enable Atlas backups. Pro users get encrypted cloud backup in-app.

11. Success Metrics (KPIs)

Metric	Month 3 Target	Month 6 Target	Month 12 Target
Total Installs	500	5,000	50,000
Daily Active Users (DAU)	100	1,000	10,000
Free to Pro Conversion	2%	4%	6%
Monthly Revenue	Rs. 0	Rs. 25,000	Rs. 2,50,000
App Store Rating	4.0+	4.2+	4.4+
Voice Extractions/Day	300	3,000	30,000
Monthly Churn (Pro)	-	<10%	<7%
Crash-free Sessions	98%	99%	99.5%

12. Immediate Next Steps

Start Today: These 3 actions should be completed before any code is written

Install development tools: Node.js, Python 3.11, Android Studio, Xcode via Mac App Store

Create accounts: MongoDB Atlas (free), OpenAI Platform, Render.com, Razorpay

Buy domain: vocash.in or preferred name on GoDaddy/Namecheap

This Week: Complete Phase 1 — working FastAPI backend with AI extraction running locally

Set up FastAPI project with folder structure

Connect MongoDB Atlas free cluster

Build user auth endpoints (register + login + JWT)

Integrate OpenAI GPT-4o-mini extraction with Hindi/English prompt

Build daily usage limit checker (3 free calls per day)

Test all APIs in Swagger UI at localhost:5000/docs

Next Week: Phase 2 — React Native app with voice input working on Android + iOS Simulator

VOCASH — Build Fast. Launch Smart. Grow Organically.

"Bas Bolo, Hum Sunenge"